

Literature Survey

Automated AI-based PCBA's Inspection

1. Introduction

Automated inspection of PCBAs has become essential due to increasing component density and miniaturization. Manual inspection is inconsistent and error-prone, which led to the development of automated visual inspection systems. Early research focused on rule-based image processing methods, while recent advancements emphasize AI-driven approaches for higher accuracy and adaptability.

2. Industry Standard Inspection Techniques

PCBAs are commonly inspected using Automated Optical Inspection (AOI), Automated X-ray Inspection (AXI), Solder Paste Inspection (SPI), and In-Circuit Testing (ICT). AOI detects visible surface defects, while AXI identifies hidden solder defects in components such as BGAs. These inspection methods follow IPC-A-610 standards, which define defect acceptability criteria in electronics manufacturing. Modern industrial systems integrate 3D imaging and AI-based classification to reduce false calls.

3. Conventional Image Processing Techniques

Traditional PCB inspection systems relied on thresholding, edge detection, morphological operations, contour analysis, and template matching. As highlighted in classical surveys, these methods were computationally efficient but sensitive to lighting variations and complex board layouts. Their rule-based nature limited adaptability to new defect types.

4. AI-Based Inspection Techniques

Recent research applies Convolutional Neural Networks (CNNs) and object detection models such as YOLO and Faster R-CNN for defect localization and classification. Deep learning methods improve accuracy, reduce false positives,

and adapt better to complex PCB designs. Transfer learning is often used to address limited defect datasets.

5. Competitive and Industrial Research

Leading manufacturers such as Koh Young and ViTrox have incorporated AI into 3D AOI systems to enhance solder joint inspection and reduce manual verification. Current industry trends focus on smart factory integration, real-time analytics, and predictive quality control under Industry 4.0.

6. Comparative Analysis

Conventional rule-based systems are simpler and computationally lighter but lack flexibility. AI-based inspection systems require more computational resources and training data but provide significantly better accuracy and robustness in complex PCB assemblies.

7. Research Challenges

Key challenges include high system cost, limited labeled datasets, variability in lighting and board designs, and real-time processing constraints. Research continues toward lightweight AI models and hybrid inspection systems.

References

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